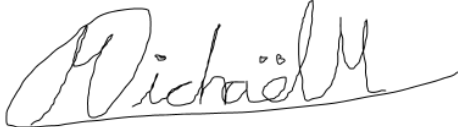


## LAB 6: Time Response Analysis of Dynamic Systems

|               |                              |
|---------------|------------------------------|
| Course Number | MENG 3020                    |
| Course Title  | System Modeling & Simulation |
| Semester/Year | Fall 2025                    |

|                         |                                           |
|-------------------------|-------------------------------------------|
| Lab/Tutorial Report No. | Lab 6                                     |
| Report Title            | Time Response Analysis of Dynamic Systems |
| Section No.             | 0NA                                       |
| Group No.               | N/A                                       |
| Submission Date         | November 16 <sup>th</sup> , 2025          |
| Due Date                | November 17 <sup>th</sup> , 2025          |

| Student Name      | Signature*                                                                           | Total Mark |
|-------------------|--------------------------------------------------------------------------------------|------------|
| Michaël McCorkell |  | / 50       |

\* By signing above, you attest that you have contributed to this submission and confirm that all work you have contributed to this submission is your work. Any suspicion of copying or plagiarism in this work will result in an investigation of Academic Misconduct and may result in a ZERO on the work or possibly more severe penalties.

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**LAB 6 Grading Sheet**

|                                                                                     |                                 |
|-------------------------------------------------------------------------------------|---------------------------------|
| Student First Name:<br>Michaël                                                      | Student Last Name:<br>McCorkell |
| Part A: Analysis of First-Order Transfer Function Models                            | /10                             |
| PART B: Transfer Function Model of Nonlinear Systems from Step Response             | /10                             |
| Part C: Analysis of Second-Order Transfer Function Models                           | /15                             |
| Post Lab Assignment                                                                 | /10                             |
| General Formatting: Clarity, Writing style, Grammar, Spelling, Layout of the report | /5                              |
| Total Mark                                                                          | /50                             |

# LAB 6: Time Response Analysis of Dynamic Systems

## OBJECTIVES

- To understand some of the basic concepts behind the dynamic systems analysis, such as stability, time response characteristics, nonlinearity, and linear models.
- To explore step response of first-order and second-order systems and derive a first-order transfer function model of a nonlinear dynamic system from step response, and study the limitations of nonlinear systems modeling.

## INTRODUCTION

### POLES and ZEROES

Recall that the **poles** of a transfer function model are those values of the Laplace transform variable,  $s$ , for which the denominator of the transfer function is zero, including any roots shared with the numerator, i.e., roots of the denominator which 'cancel' out with the numerator. Similarly, the zeros of a transfer function are those values of  $s$  that cause the transfer function to become zero, including roots shared with the denominator.

### STABILITY of LTI SYSTEMS

The time response of a linear time invariant (LTI) dynamic system  $y(t) = y_{forced}(t) + y_{free}(t)$ , where  $y_{forced}(t)$  is the **forced response** (driven by the input applied to the system) and  $y_{free}(t)$  is the **free response** (driven by the system's initial states).

A linear, time-invariant system is called:

- stable** if the free response approaches zero as time approaches infinity.
- unstable** if the free response grows without bound as time approaches infinity.
- marginally stable** if the free response neither decays nor grows but remains constant or oscillates as time approaches infinity.

For LTI dynamic systems one can discuss stability easily in terms of the locations of the poles of the system's transfer function. A system is **stable** if all poles lie in the **left-half** of the complex  $s$ -plane (LHP). A system is **unstable** if its transfer function has at least one pole in the **right-half** of the complex plane (RHP), or a pole of multiplicity greater than one on the imaginary axis. A system is **marginally stable** if it has no poles in the RHP and only poles of multiplicity one on the **imaginary axis**.

### FIRST-ORDER and SECOND-ORDER TRANSFER FUNCTION MODELS

**First-order** transfer function models has the following general form,

$$G(s) = \frac{K}{\tau s + 1}$$

where,  $K$  : is the steady-state gain,  
 $\tau$  : is the time-constant,

The **steady-state gain** affects the final value of the step response, and the **time-constant** represents how fast the system response reaches the final value.

**Second-order** transfer function models has the following general form,

$$G(s) = \frac{K \omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2}$$

where,  $K$  : is the steady-state gain,

$\zeta$  : is the damping ratio,

$\omega_n$  : is the natural undamped frequency,

The **steady-state gain** affects the final value of the step response, and the **damping ratio** and the **natural undamped frequency** determined the speed and stability of the response.

1. Under-damped Systems:  $0 < \zeta < 1$
2. Critically-damped Systems:  $\zeta = 1$
3. Over-damped Systems:  $\zeta > 1$
4. Undamped Systems:  $\zeta = 0$

## Part A: Analysis of First-Order Transfer Function Models

Consider the following transfer function models.

$$G_1(s) = \frac{0.5}{s+0.5}, \quad G_2(s) = \frac{s+1}{0.5s+1}, \quad G_3(s) = \frac{s+3}{s+0.3}, \quad G_4(s) = \frac{1}{s-1}$$

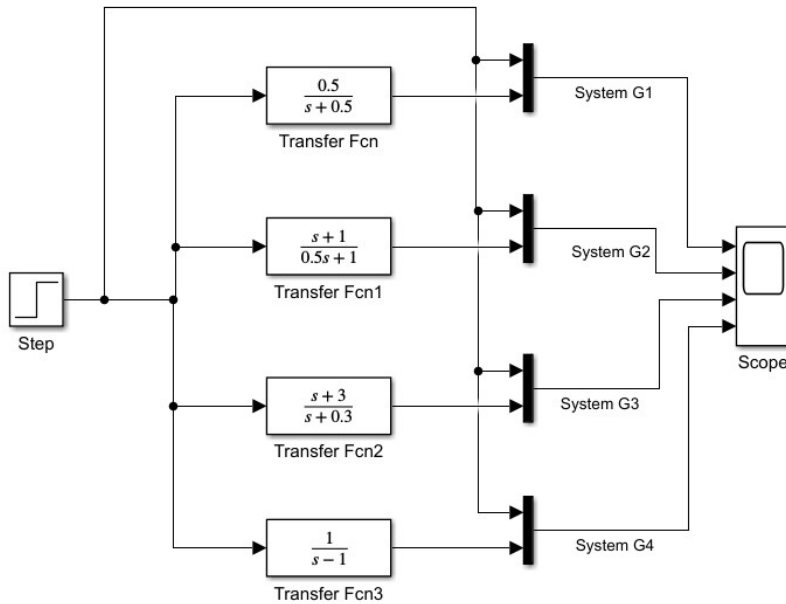
Complete the required information in **Table 1** for each transfer function:

**Table 1**

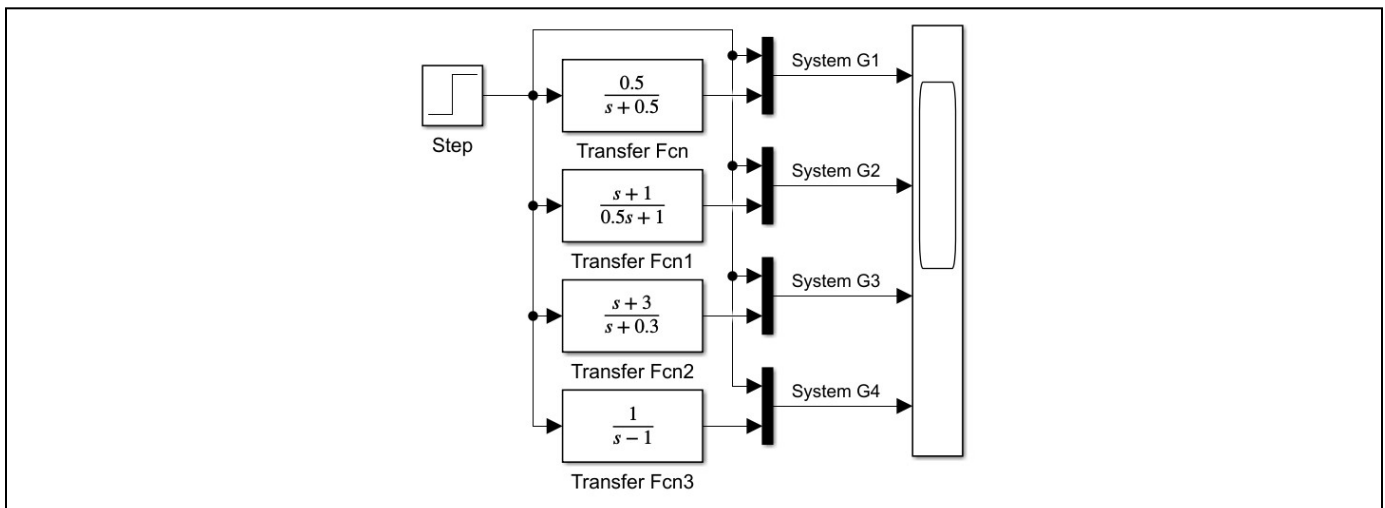
| Transfer Function | Poles | Zeros | Initial Value of Step Response | Steady-State Value of Step Response | Stability (YES/NO) | Time Constant |
|-------------------|-------|-------|--------------------------------|-------------------------------------|--------------------|---------------|
| $G_1(s)$          | -0.5  | None  | 0                              | 1                                   | Yes                | 2             |
| $G_2(s)$          | -10   | -1    | 2                              | 1                                   | Yes                | 0.1           |
| $G_3(s)$          | -0.3  | -3    | 1                              | 10                                  | Yes                | 3.33          |
| $G_4(s)$          | 1     | None  | 0                              | Undefined                           | No                 | 1             |

1. Start **Simulink** and open a new window.

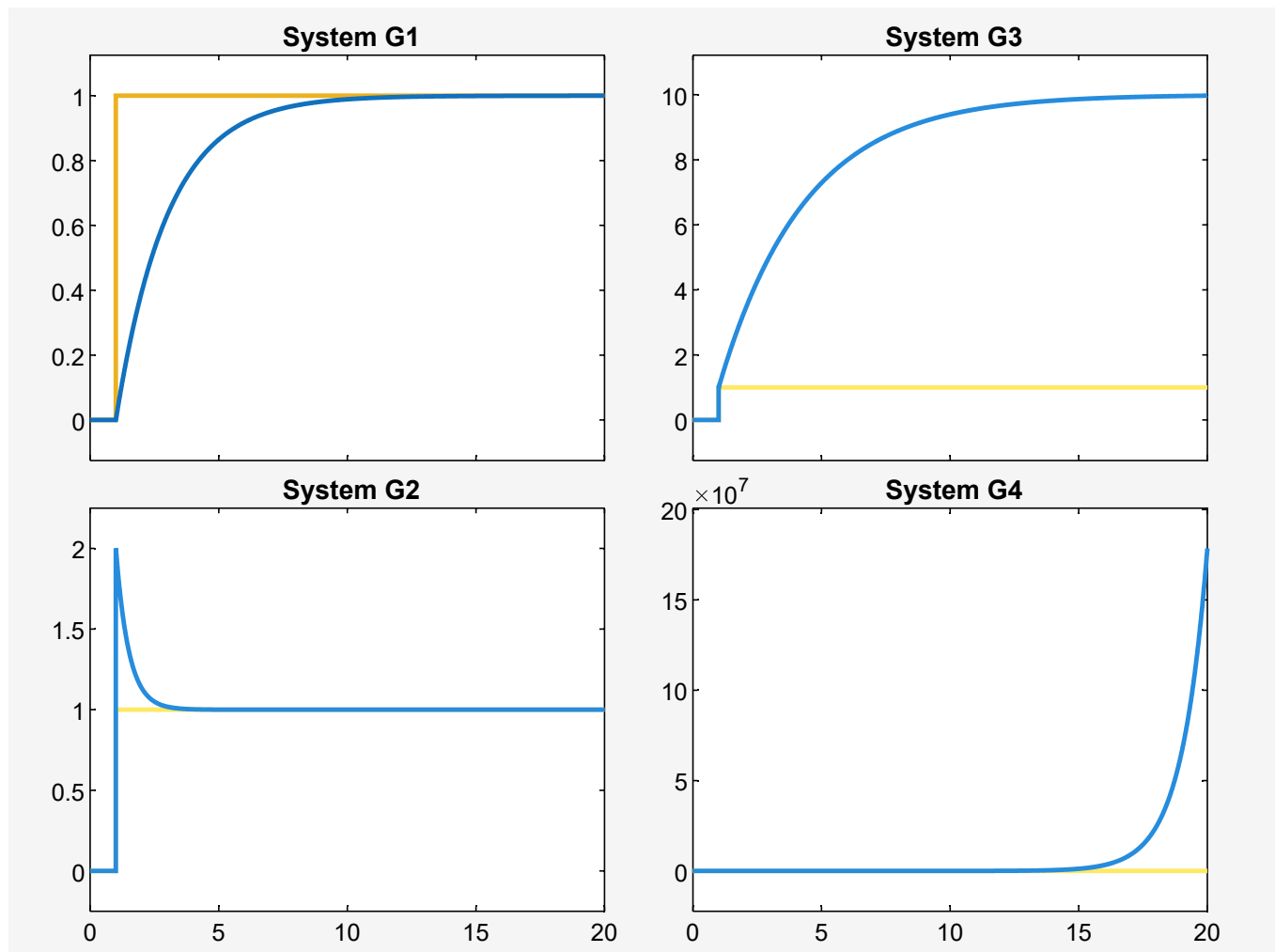
2. Create the following block diagram in **Simulink**:



3. Provide your **Simulink** model diagram below:



- Click on the **Model Settings** icon in the **MODELING** tab to open the **Configuration Parameters** window, click on the **Solver details**. Set the **Relative tolerance** value to **1e-12** to increase the resolution of the simulation. Then click **OK**.
- Save** the model file as **Lab6\_PartA.slx**.
- Set the **Step Time** of the **Step** block to **0**. Set the simulation **Stop Time** **20 sec** and **Run** the simulation. Open the **Scope** block. You will see the **input-output** curves for each transfer function. Provide the plot below:



7. Compare the step response graphs with the results of **Table 1**. Are the results consistent with each other?

The step response results for the four systems are mostly consistent with the values listed in Table 1. G1(s) and G3(s) match the table perfectly in terms of initial value, steady-state value, and stable behavior. For G2(s), the plotted response shows an initial value of 2. Finally, G4(s) behaves exactly as expected for an unstable system—the response diverges rapidly—confirming that G4(s) is not stable despite the table’s description already marking it as unstable. Overall, the graphs agree with the expected system behavior after correcting the initial value of G2(s).

## PART B: Transfer Function Model of Nonlinear Systems from Step Response

Dynamic models are essential for understanding the system behavior and design control systems. These models can either be derived mathematically from the first principles of physics of process or **empirically** from the **step response** of the process. The purpose of this part is to obtain a transfer function model of the following Single-Tank system experimentally from the input- output data.

Consider the following **Single-Tank** System as shown below:

The **Single-Tank** system consists of a pump with a water basin and a tank. The pump drives the water vertically from the bottom water basin up to the top of the system. The flow from the tank flows through an outlet orifice located at the bottom of the tank into the main water reservoir.

Liquid level control is common in many industries, such as pulp and paper mills, petrochemical plants and water treatment facilities.

Parameters of the Single-Tank system are defined as follows:

- $V_p$ : Voltage to the pump ( $0 V \leq V_p \leq 50 V$ )
- $D_i$ : Tank inside diameter (30 cm)
- $D_o$ : Out flow orifice diameter (1 cm)
- $L$ : The water level in the tank

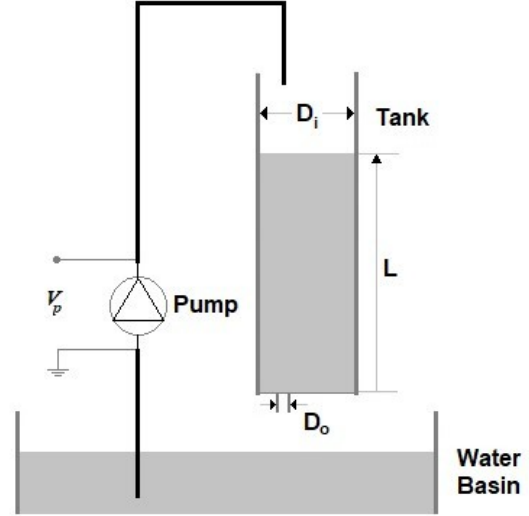


Figure 1: Single-Tank System

The rate of flow can be changed by changing the voltage of the pump ( $V_p$ ) and/or by using outflow orifices with different diameters. Here, the *voltage to the pump* ( $V_p$ ) is considered as the **input** to the process and the *water level in the tank* ( $L$ ) is the **output** of the system.

Applying the mass balance principle to the water level in the Tank, the **equation of motion** of the Tank is derived as:

$$A \frac{dL(t)}{dt} = F_i(t) - F_o(t)$$

where,  $A$  is the **area of the Tank**,

$$A = \frac{\pi D_i^2}{4}$$

$F_i$  and  $F_o$  are the **inflow rates** and **outflow rates** of the Tank, respectively. Assume that the volumetric **inflow rate** into the Tank is directly proportional to the applied pump voltage:

$$F_i(t) = K_p V_p(t)$$

where,  $K_p$  is the **pump flow constant**. Applying the Bernoulli's equation for small orifices, the **outflow velocity** from Tank can be expressed by the following relationship,

$$v_o(t) = \sqrt{2gL(t)}$$

where,  $g$  is the **gravitational acceleration**. Lastly, the outflow rate from Tank can be expressed as below:

$$F_o(t) = A_o v_o(t)$$

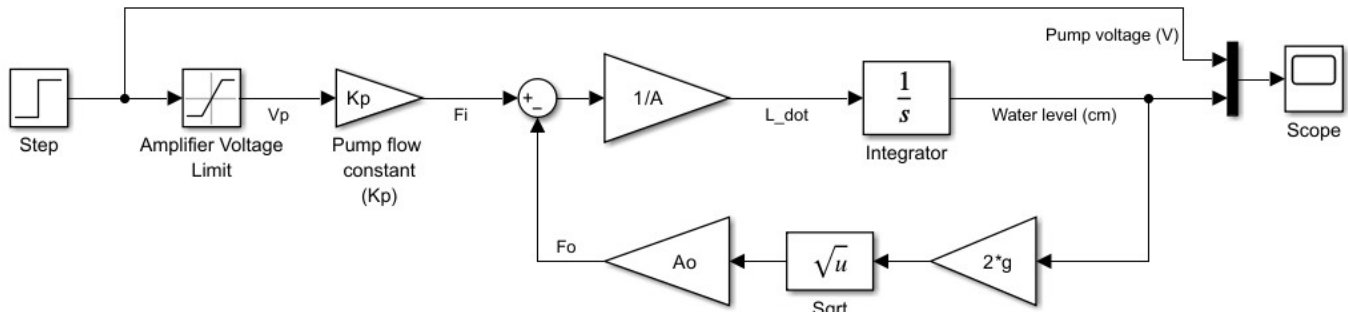
where,  $A_o$  is the **outlet area of Tank**.

$$A_o = \frac{\pi D_o^2}{4}$$

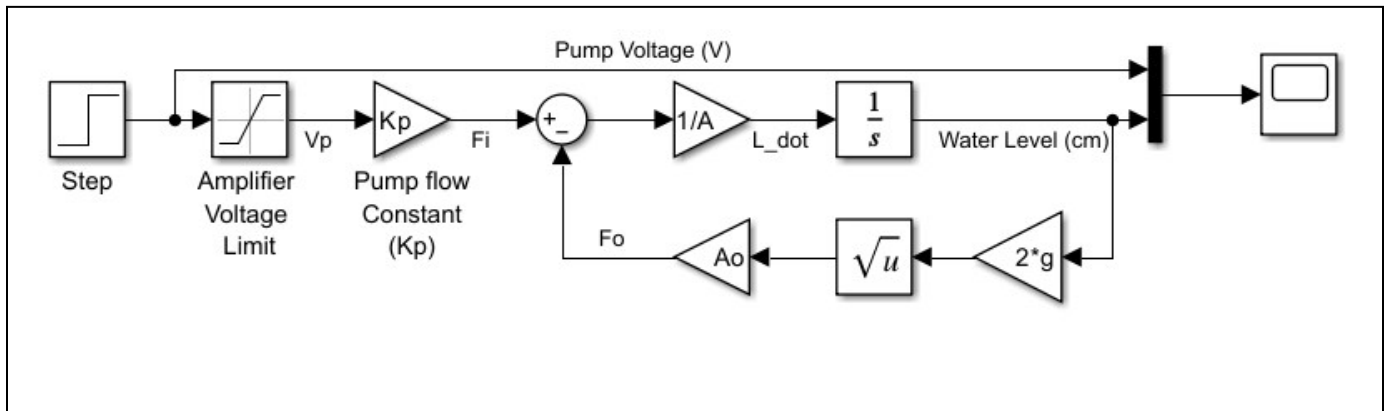
Therefore, the **nonlinear** differential equation of dynamic system is obtained as:

$$A \frac{dL(t)}{dt} = K_p V_p(t) - A_o \sqrt{2gL(t)}$$

8. Open a new **Simulink** model window.
9. Create the following nonlinear model of the Single-Tank system in **Simulink**.



10. Provide your **Simulink** model diagram below:

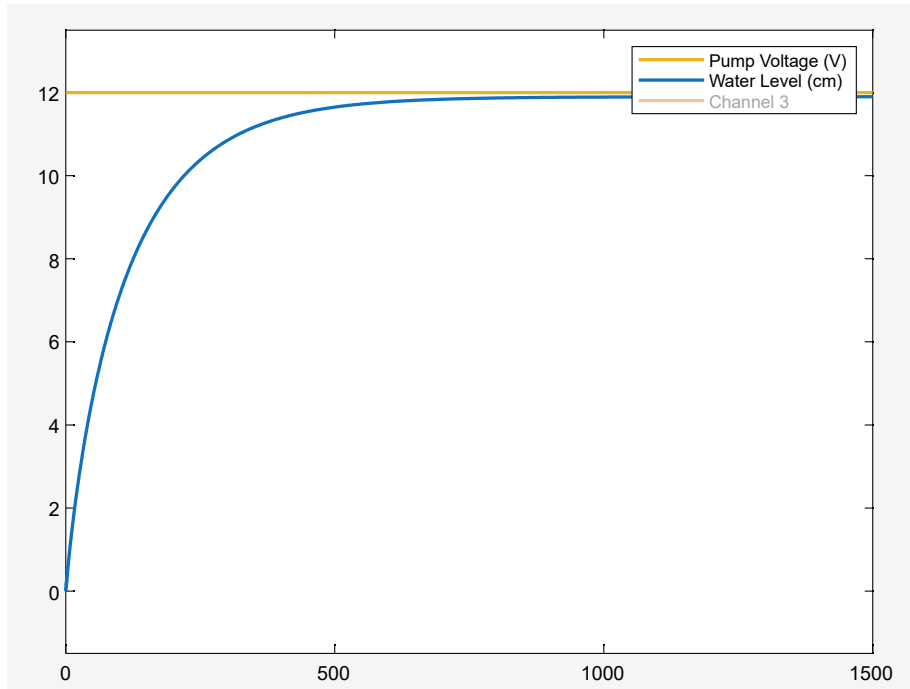


11. Click on the **Model Settings** icon in the **MODELING** tab to open the **Configuration Parameters** window, click on the **Solver details**. Set the **Relative tolerance** value to **1e-12** to increase the resolution of the simulation. Then click **OK**.
12. **Save** the model file as **Lab6\_PartB.slx**.
13. In the **Saturation** block, set the pump voltage limits as  $0V \leq V_p \leq 50V$ .
14. In the **Step** block, set **Step Time** to **0** and apply the **Final value** of **12V** as the pump voltage.
15. **Right-click** in the **Simulink** model window, select **Model Properties** from drop down menu. This will bring up the **Model Properties** dialog box.
16. Select the **Callbacks** tab.
17. Select **InitFcn** from the list of **Model callbacks**.
18. Type the following **MATLAB** commands into the panel under **Model initialization function**. Click **Ok**. These commands will be executed at the start of the model simulation.

```
Di = 30;           % Tank inside diameter [cm]
Do = 1;           % Out flow orifice diameter [cm]
A = pi*Di^2/4;    % Area of the Tank [cm^2]
```

```
Ao = pi*Do^2/4;      % Outlet area of Tank [cm^2]
g = 981;             % Gravitational acceleration [cm/s^2]
Kp = 10;             % Pump flow constant [cm^3/V.s]
```

19. Set the simulation **Stop Time 1500 sec** and, **Run** the simulation. Open the **Scope** block. You will see the **pump voltage** and the **water level** curves. Provide the plot below:



20. Activate the **Cursor Measurements** of **Scope** and find the **DC-gain** and the **time constant** from the step response plot. Insert the required values in the first row of **Table 2**.

21. Determine a **first-order** transfer function model for the system. Provide the transfer function model in the standard form as:

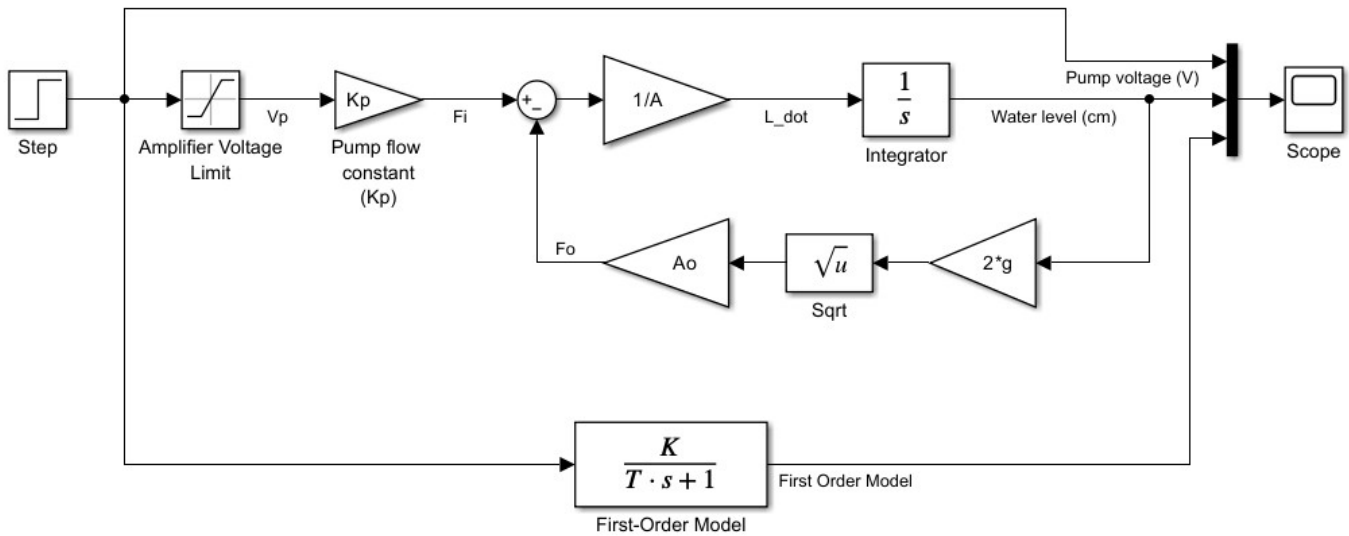
$$G(s) = \frac{K}{\tau s + 1}$$

Complete the first row of **Table 2** based on your calculations.

**Table 2**

| Applied Voltage (V) | Water Level (cm) | Time Constant (sec) | DC-gain (cm/V) | Transfer Function Model G(s)       |
|---------------------|------------------|---------------------|----------------|------------------------------------|
| <b>12 V</b>         | 11.9 cm          | 110 sec             | 0.99167        | $G(s) = \frac{0.99167}{110s + 1}$  |
| <b>15 V</b>         | 18.59 cm         | 137.5 sec           | 1.2393         | $G(s) = \frac{1.2393}{137.5s + 1}$ |
| <b>18 V</b>         | 26.75 cm         | 164.7 sec           | 1.4861         | $G(s) = \frac{1.4861}{164.7s + 1}$ |

22. Create the first-order transfer function model in **Step 21** and add it to your **Simulink** model as shown below:

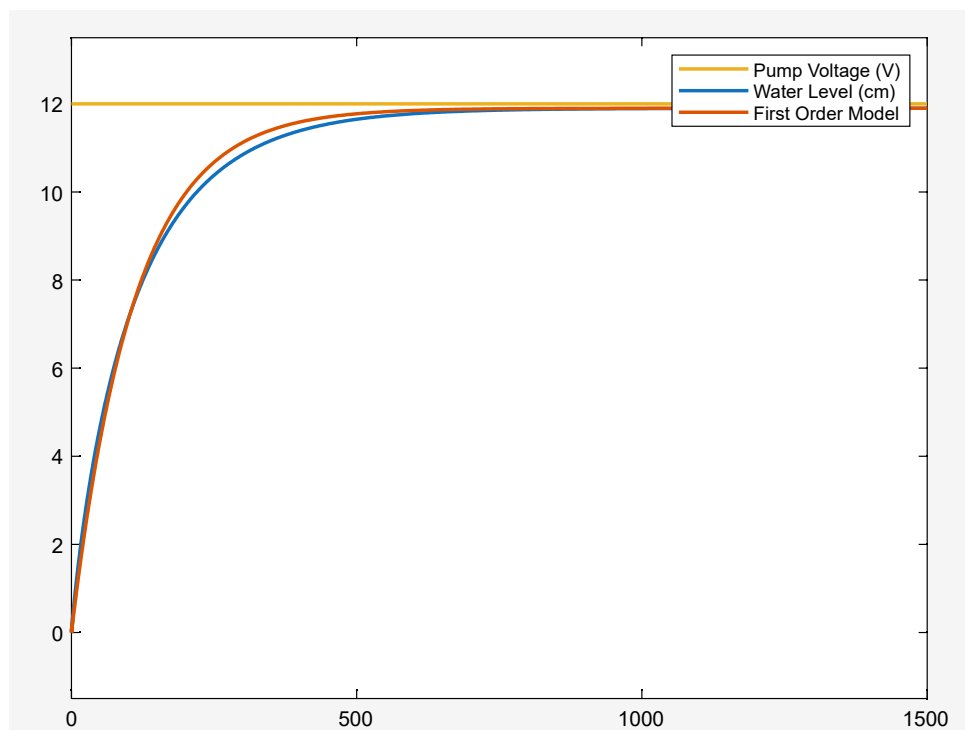


23. Run the simulation. Open the **Scope** block. Compare the step response of the nonlinear system and the linear first-order transfer function model. Is the linear model acceptable?

The linear first-order model appears to be a reasonable approximation, particularly for steady-state analysis and cases where an approximate transient reaction is adequate. However, because of its inability to accurately represent nonlinear dynamics, the linear model may introduce modest flaws for very precise transient response analysis. Overall, the linear model may be deemed sufficiently accurate for a variety of practical applications.

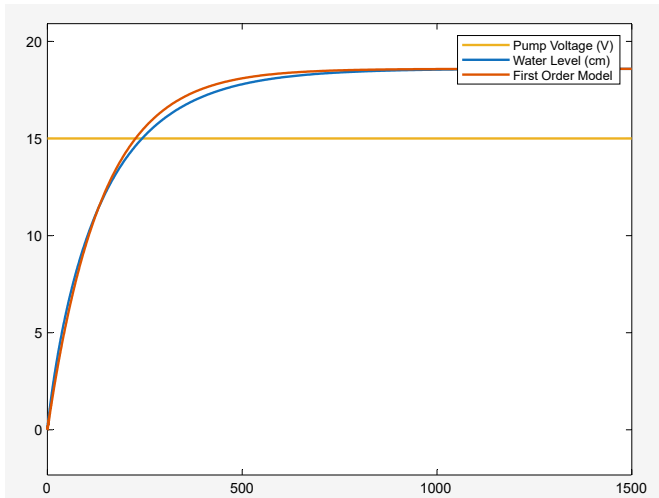
Provide the plot below:

**Pump Voltage = 12V & Water Level (cm)**

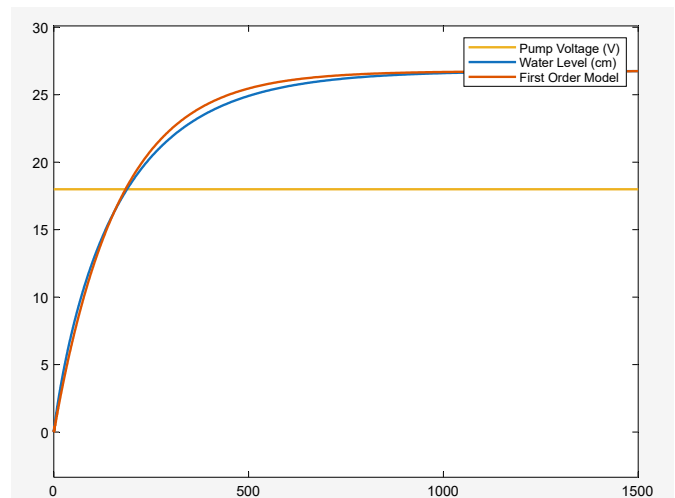


24. Change the **applied voltage** according to the provided values in **Table 2**. Plot the graphs and provide them below.

**Pump Voltage = 15V & Water Level (cm)**



**Pump Voltage = 18V & Water Level (cm)**



25. Find the **DC-gain**, the **time constant** and the **transfer function model** for each system and complete the **Table 2**.
26. Compare the obtained linear transfer function models in **Table 2** for each operation point. Explain effect of changing the operating point of the nonlinear Single-Tank system on the linear model. Can we use a single model for all operation points?

The change in both time constant and DC gain shows that the system's dynamics vary with the operating point. As the water level increases, the system:

- Responds more slowly (higher time constant), likely due to the increased resistance to further increases in water level as the tank fills.
- Becomes more sensitive to input changes (higher DC gain), meaning that a small change in voltage has a larger impact on the water level at higher points.

This variability in the model parameters is characteristic of nonlinear systems, where the system behavior depends on the operating point.

In conclusion, a single linear model is insufficient for accurately representing the system's behavior across all operating points. Instead, multiple linear models or a nonlinear model would better capture the system's dynamics across different voltage levels and water heights.

## Part C: Analysis of Second-Order Transfer Function Models

Consider the series RLC circuit as shown in Figure 2,

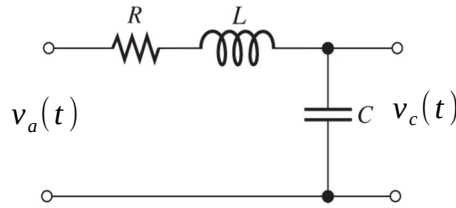


Figure 2: Series RLC Circuit

The **dynamic model** of this system is described by the following ordinary differential equation,

$$v_a(t) = LC \frac{d^2 v_c(t)}{dt^2} + RC \frac{dv_c(t)}{dt} + v_c(t)$$

where  $R$ : Resistance ( $\Omega$ ),  
 $C$ : Capacitance (F),  
 $L$ : Inductance (H),  
 $v_a(t)$ : Applied voltage (V),  
 $v_c(t)$ : Voltage across the capacitor (V),

Considering the  $v_a(t)$  as the input and  $v_c(t)$  as the output, the **transfer function model** is obtained as below,

$$G(s) = \frac{V_c(s)}{V_a(s)} = \frac{1}{LC s^2 + RCs + 1}$$

27. Find the transfer function model in the **standard** second-order transfer function form and determine the **DC-gain**  $K$ , **damping ratio**  $\zeta$  and the **undamped natural frequency**  $\omega_n$  in terms of the  $R$ ,  $C$  and  $L$  values and complete **Table 3**. Show your work below.

$$\text{Standard Form} \quad \rightarrow \quad \frac{K \omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2}$$

$$G(s) = \frac{1}{LCs^2 + RCs + 1} \rightarrow LCs^2 + RCs + 1 = s^2 + \frac{R}{L}s + \frac{1}{LC}$$

Transfer Function in standard second-order form is

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \rightarrow \frac{\frac{1}{LC}}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

DC-gain  $K=1$

$$\text{Damping Ratio } \zeta: \frac{R}{2} \sqrt{\frac{C}{L}}$$

$$\text{Undamped Natural Frequency } \omega_n: \sqrt{\frac{1}{LC}}$$

Table 3

| Transfer Function Model            | DC – gain $K$ | Damping Ratio $\zeta$            | Undamped Natural Frequency $\omega_n$ |
|------------------------------------|---------------|----------------------------------|---------------------------------------|
| $G(s) = \frac{1}{LCs^2 + RCs + 1}$ | 1             | $\frac{R}{2} \sqrt{\frac{C}{L}}$ | $\sqrt{\frac{1}{LC}}$                 |

28. Assume that  $R=100 \ \Omega$ ,  $C=0.05 \ F$ ,  $L=5 \ H$ . Open a new **script** file and run the following code in **MATLAB** to create the transfer function model of the system.

```

R = 100;           % Resistance (Ohms)
C = 0.05;          % Capacitance (F)
L = 5;             % Inductance (H)

num = [1/(L*C)];   % Numerator coefficients
den = [1 R/L 1/(L*C)]; % Denominator coefficients
G = tf(num,den)    % Create transfer function

z = R*sqrt(C/L)/2  % Damping Ratio
wn = 1/sqrt(L*C)   % Undamped Natural Frequency (rad/sec)

P = pole(G)        % Poles

figure; pzmap(G)    % Pole-Zero Map
figure; step(G)     % Step response plot
xlim([0 30])        % Time axis limit of step response plot

stepinfo(G)         % Step response specifications

```

Provide the results from **MATLAB Command** window and complete the first row of **Table 4**:

$$G = \frac{4}{s^2 + 20s + 4} \quad P \begin{Bmatrix} -19.7980 \\ -0.2020 \end{Bmatrix}$$

$$z = 5$$

$$\omega_n = 2$$

Rise Time: 10.8747

Transient Time: 19.4140

Settling Time: 19.4140

Settling Min: 0.9035

Settling Max: 0.9993

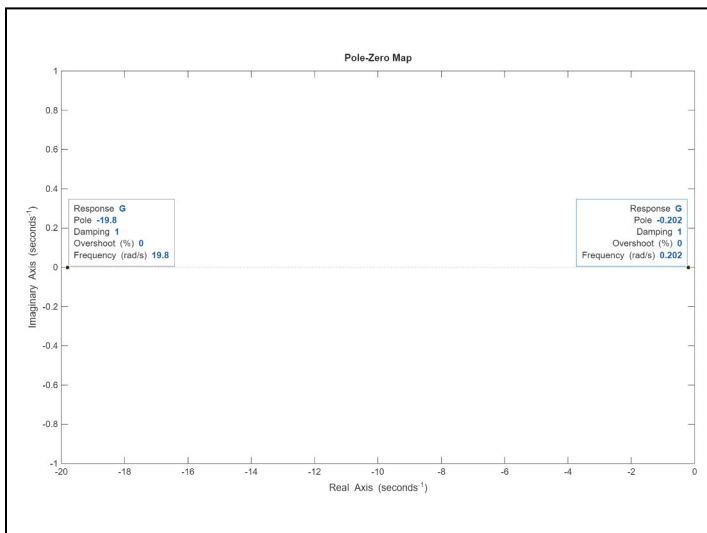
Overshoot: 0

Undershoot: 0

Peak: 0.9993

Peak Time: 36.2413

Pole-Zero Map (R = 100Ω)



Step Response Plot (R = 100Ω)

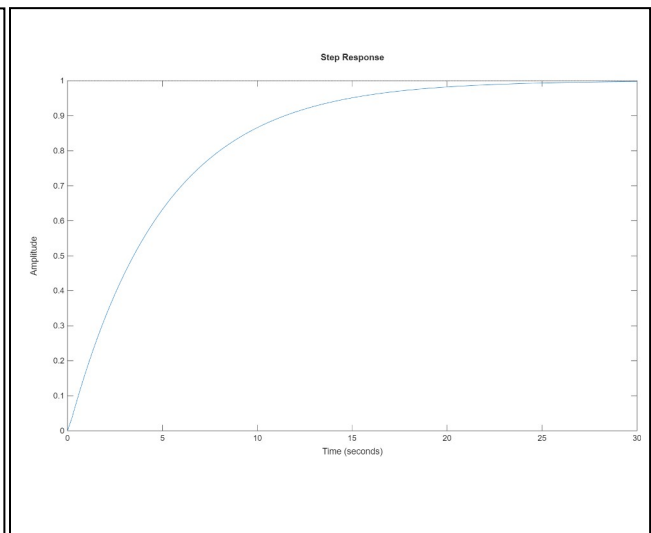


Table 4

| Resistance (Ohms) | Poles                        | Damping Ratio $\zeta$ | Undamped Natural Frequency $\omega_n$ | System damping type | Rise Time | Percent of Overshoot | Settling Time |
|-------------------|------------------------------|-----------------------|---------------------------------------|---------------------|-----------|----------------------|---------------|
| 100 $\Omega$      | -19.8<br>-0.202              | 5                     | 2                                     | Overdamping         | 10.8747   | 0                    | 19.4140       |
| 50 $\Omega$       | -9.5826<br>-0.4174           | 2.5                   | 2                                     | Overdamping         | 5.2678    | 0                    | 9.4787        |
| 20 $\Omega$       | -2<br>-2                     | 1                     | 2                                     | Overdamping         | 1.6790    | 0                    | 2.9170        |
| 10 $\Omega$       | -1+1.7321i<br>-1-1.7321i     | 0.5                   | 2                                     | Underdamped         | 0.8195    | 16.2929              | 4.0379        |
| 5 $\Omega$        | -0.5+1.9365i<br>-0.5-1.9365i | 0.25                  | 2                                     | Underdamped         | 0.6343    | 44.3235              | 7.0579        |
| 0 $\Omega$        | 0+2i<br>0-2i                 | 0                     | 2                                     | Undamped            | NaN       | NaN                  | NaN           |

29. Change the **Resistance** according to the given values in **Table 4**. Provide the required graphs, calculations, and data from the simulation and complete **Table 4**.

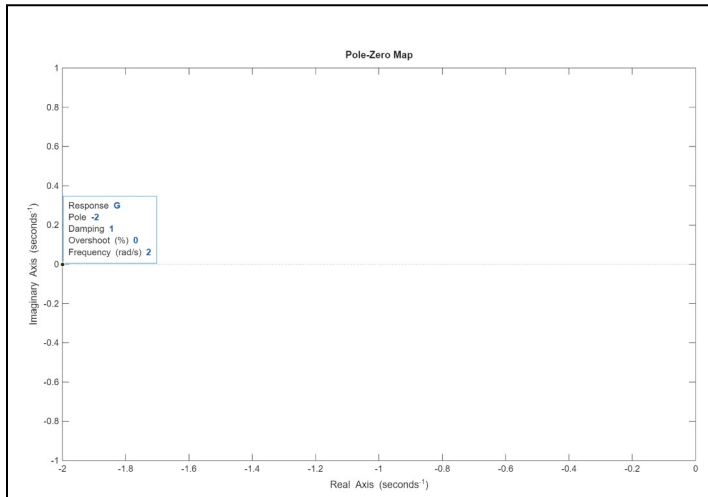
30. Explain the effect of changing the **Resistance** on the damping ratio and the natural frequency of the system.

Increasing resistance results in higher damping ratios, making the system more overdamped and reducing oscillations.

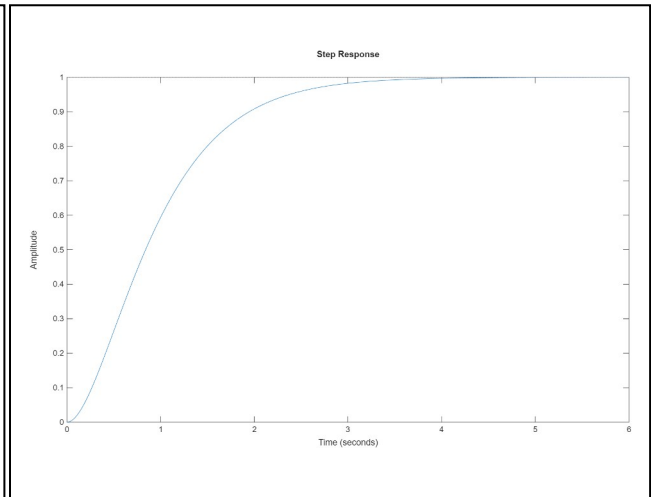
Decreasing resistance lowers the damping ratio, moving the system toward underdamped behavior with more oscillations and higher overshoot.

Natural frequency remains unchanged at  $\omega_n = 2$  for all resistance values, indicating that it is independent of the resistance in this setup.

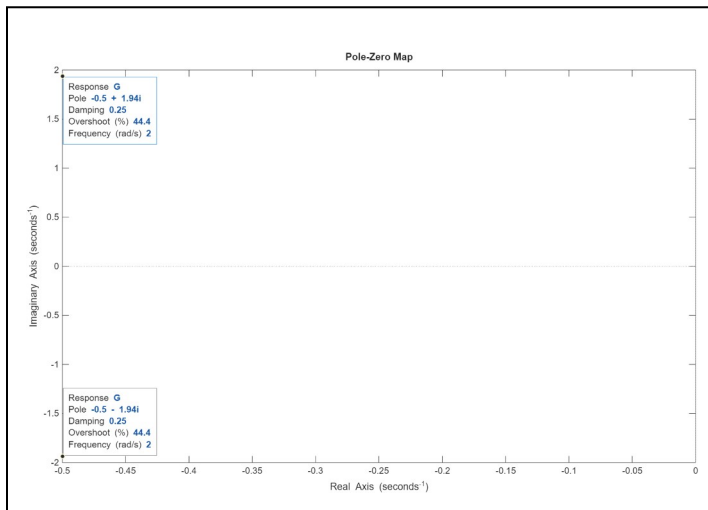
Pole-Zero Map ( $R = 20\Omega$ )



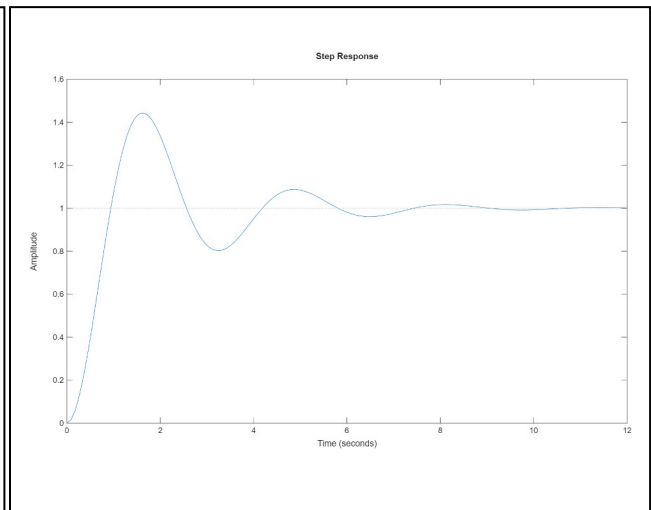
Step Response Plot ( $R = 20\Omega$ )



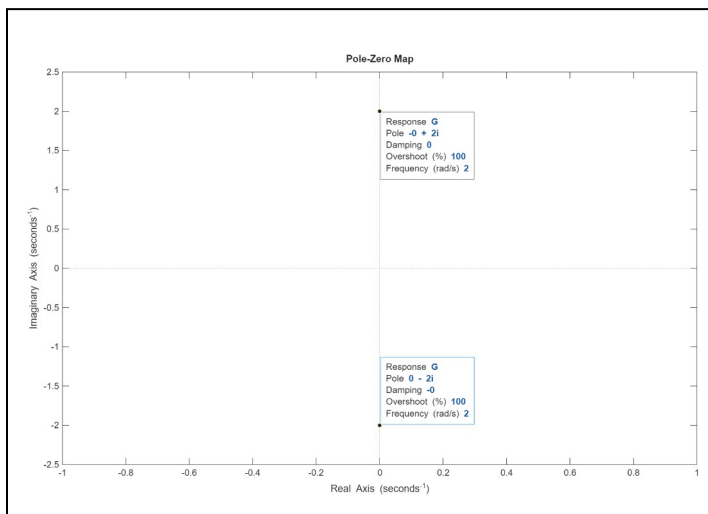
Pole-Zero Map ( $R = 5\Omega$ )



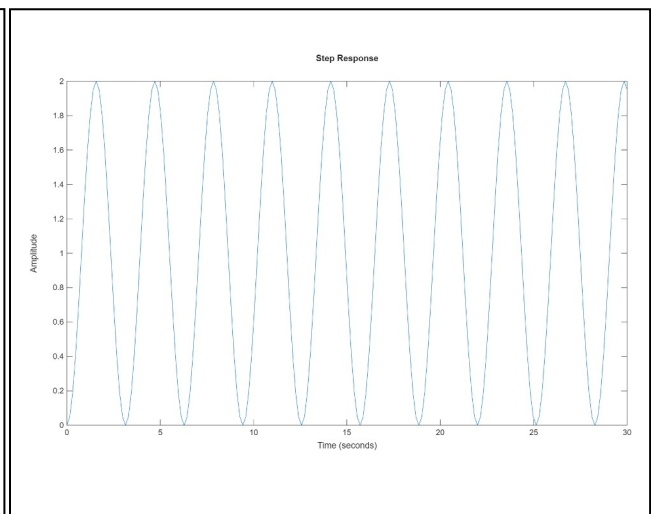
Step Response Plot ( $R = 5\Omega$ )



Pole-Zero Map ( $R = 0\Omega$ )



Step Response Plot ( $R = 0\Omega$ )



## Post Lab Assignment

1. Consider the following transfer function models:

$$G_1(s) = \frac{2.5}{s^2 + 3s + 2}, \quad G_2(s) = \frac{2.5}{s^2 + 2s + 1}, \quad G_3(s) = \frac{5}{s^2 + 4s + 13}, \quad G_4(s) = \frac{10}{s^2 + 10}$$

- Create a **MATLAB** code to obtain the transfer function and pole locations. Provide your code and the results.
  - Plot the step response and the pole-zero map of each model. Provide the graphs.
  - Determine the damping ratio and the undamped natural frequency of the models. Show your calculations.
  - Determine the damping type of each system. Over-damped, under-damped, critically-damped or undamped? Justify your answer.
2. For a system output given in the following figure and the following information, determine a second-order transfer function model. Show your calculations.

$$G(s) = \frac{K \omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2}$$

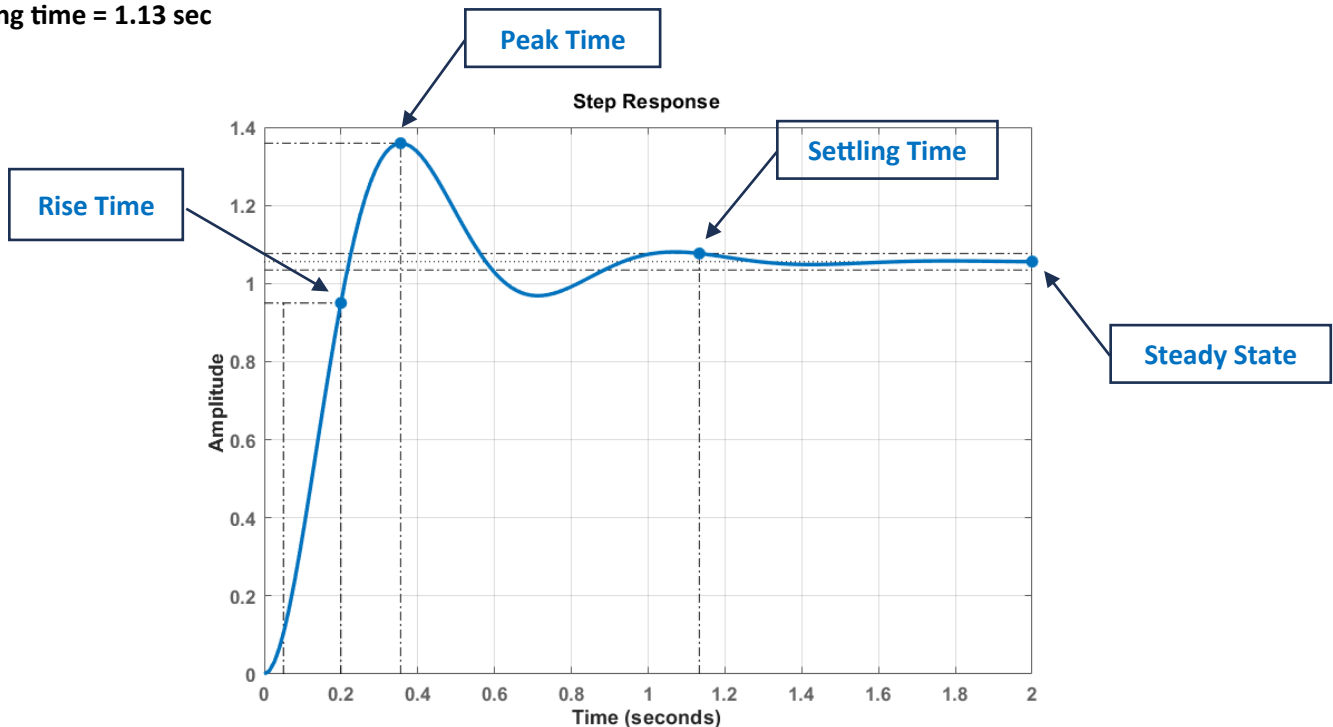
Peak time = 0.355 sec

Overshoot = 28.7%

Steady state value = 1.06

Rise time = 0.15 sec

Settling time = 1.13 sec



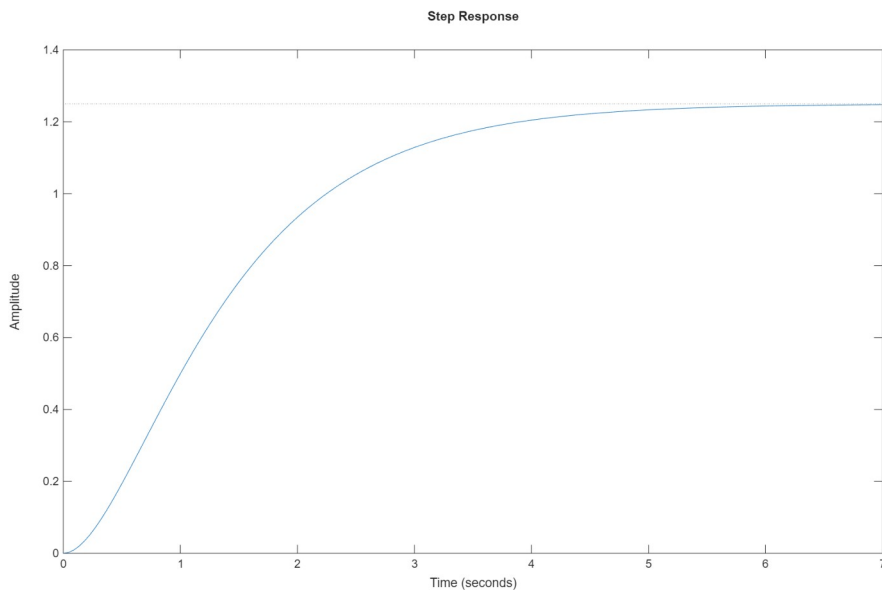
Plot the step response of the model and show the peak time, overshoot, rise-time, settling-time and steady-state value on the graph. Provide your code and the graph.

# Post Lab Assignment 1)

```

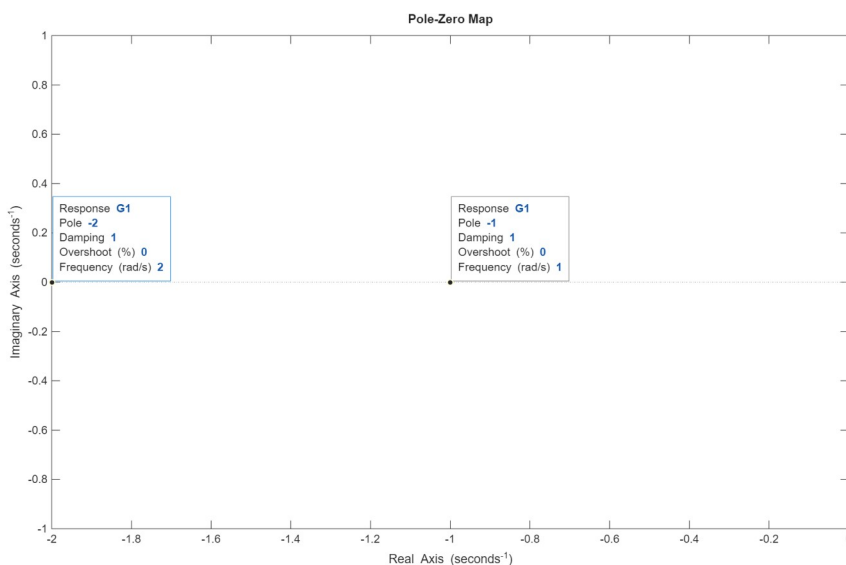
num1 = 2.5;           % Numerator coefficients
den1 = [1 3 2];       % Denominator coefficients
G1 = tf(num1,den1)     % Create transfer function
    
```

G1 =  
2.5  
-----  
s^2 + 3 s + 2



$$\begin{aligned}
 w_n &= \sqrt{2} \approx 1.414 \\
 2 \xi w_n &\rightarrow 3 \\
 \xi &= \frac{3}{2 w_n} = \frac{3}{2 * 1.414} = 1.061
 \end{aligned}$$

The damping ratio is slightly greater than 1, meaning the system is over-damped, and it will return to steady-state without oscillating but more slowly than a critically damped system



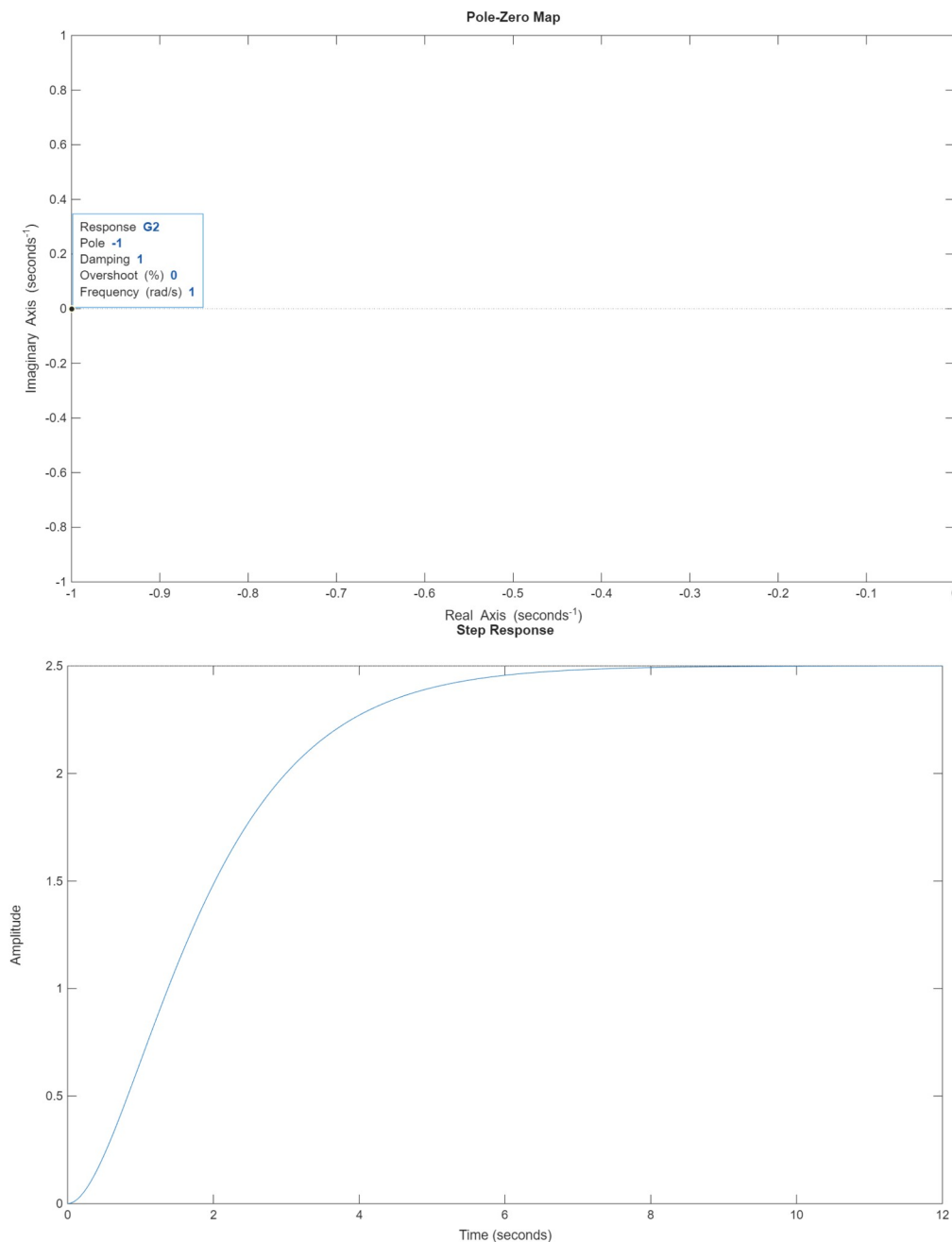
1.061 -> Overdamped

```

num2 = 2.5;           % Numerator coefficients
den2 = [1 2 1];       % Denominator coefficients
G2 = tf(num2,den2)     % Create transfer function

```

$G2 =$   
 $\frac{2.5}{s^2 + 2s + 1}$



$$\begin{aligned}
 w_n &= \sqrt{1} = 1 \\
 2\xi w_n &\rightarrow 2 \\
 \xi &= \frac{2}{2w_n} = \frac{2}{2 \cdot 1} = 1
 \end{aligned}$$

With a damping ratio of exactly 1, the system is critically damped. This is the boundary condition between oscillation and non\_oscillation, where the system returns to steady-state as quickly as possible without oscillating

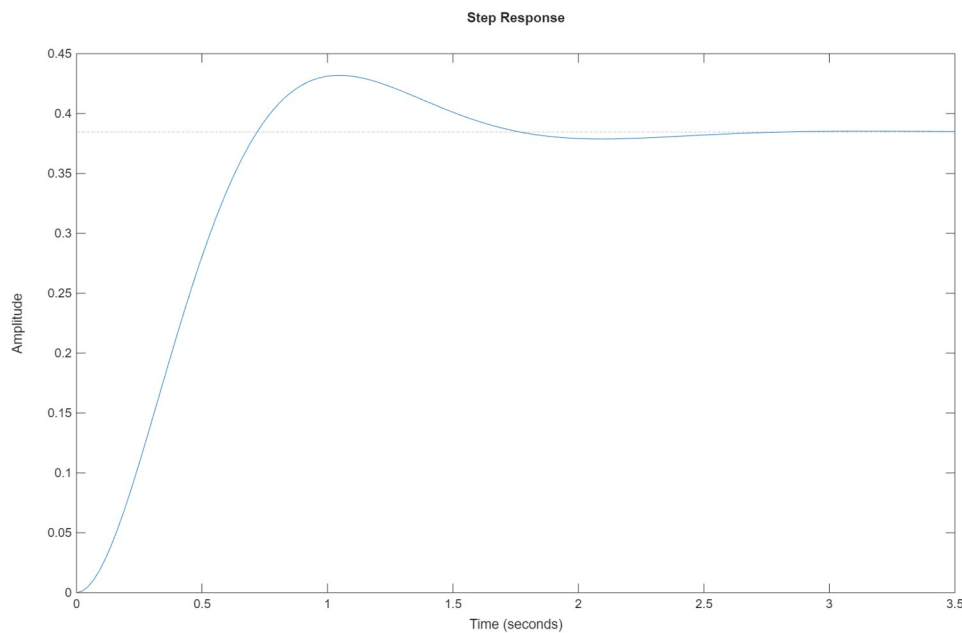
1 -> Critically Damped

```

num3 = 5;           % Numerator coefficients
den3 = [1 4 13];    % Denominator coefficients
G3 = tf(num3,den3)   % Create transfer function

```

$G3 =$   
 $\frac{5}{s^2 + 4s + 13}$

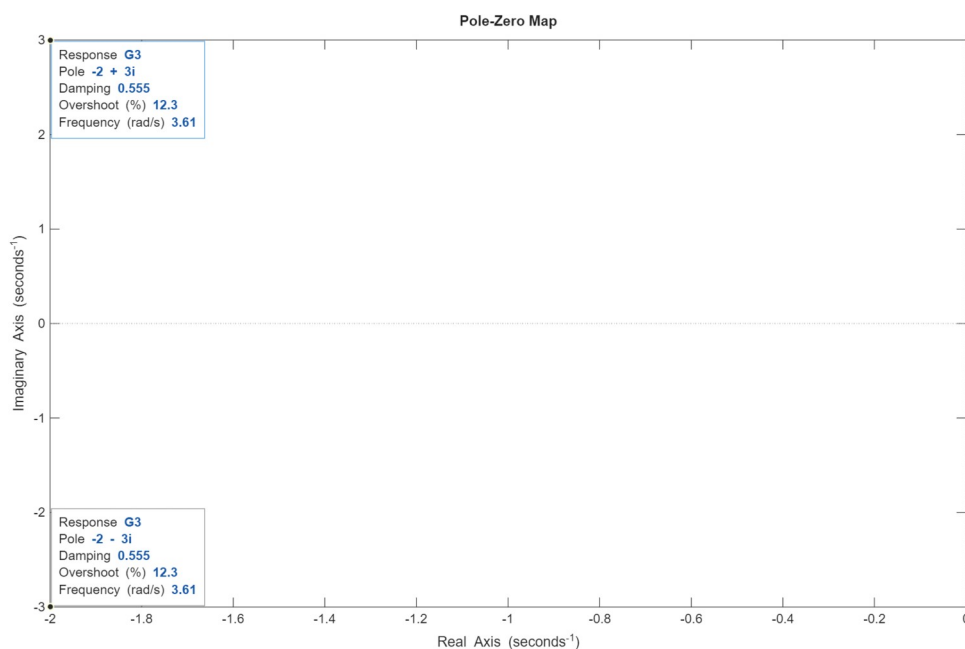


$$w_n = \sqrt{13} \approx 3.606$$

$$2\xi w_n \rightarrow 4$$

$$\xi = \frac{4}{2w_n} = \frac{4}{2 * 3.606} \approx 0.554$$

Since  $0 < \zeta < 1$ , the system is under-damped, meaning it will exhibit oscillations as it approaches steady-state



0.554 ->  
Underdamped

```

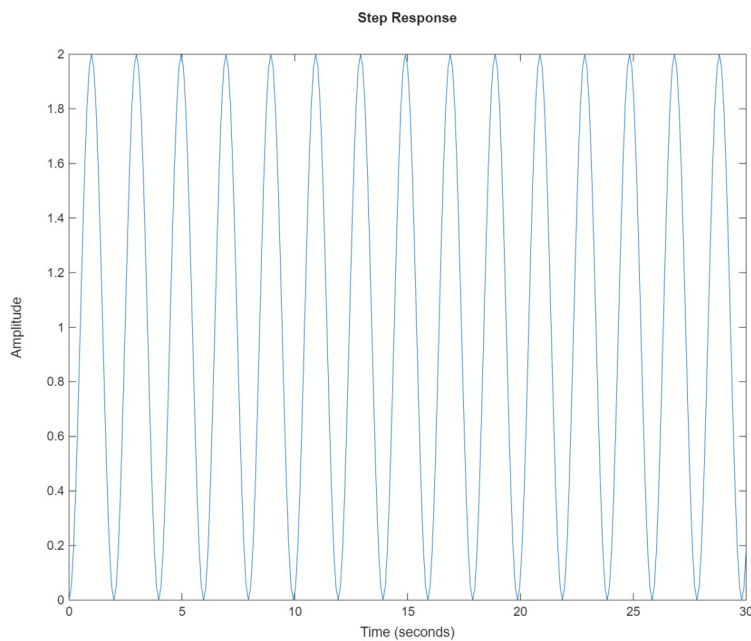
num4 = 10;           % Numerator coefficients
den4 = [1 0 10];     % Denominator coefficients
G4 = tf(num4,den4)    % Create transfer function

```

```

G4 =
    10
-----
    s^2 + 10

```

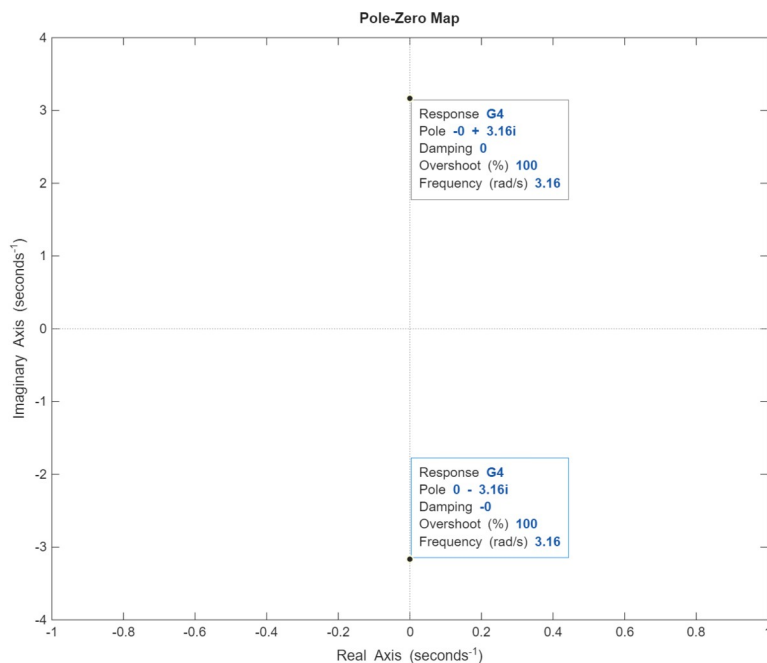


$$w_n = \sqrt{10} \approx 3.162$$

$$\xi = 0$$

With a damping ratio of 0, there is no damping in the system, and it will oscillate indefinitely without settling to a steady-state value.

0 -> Undamped



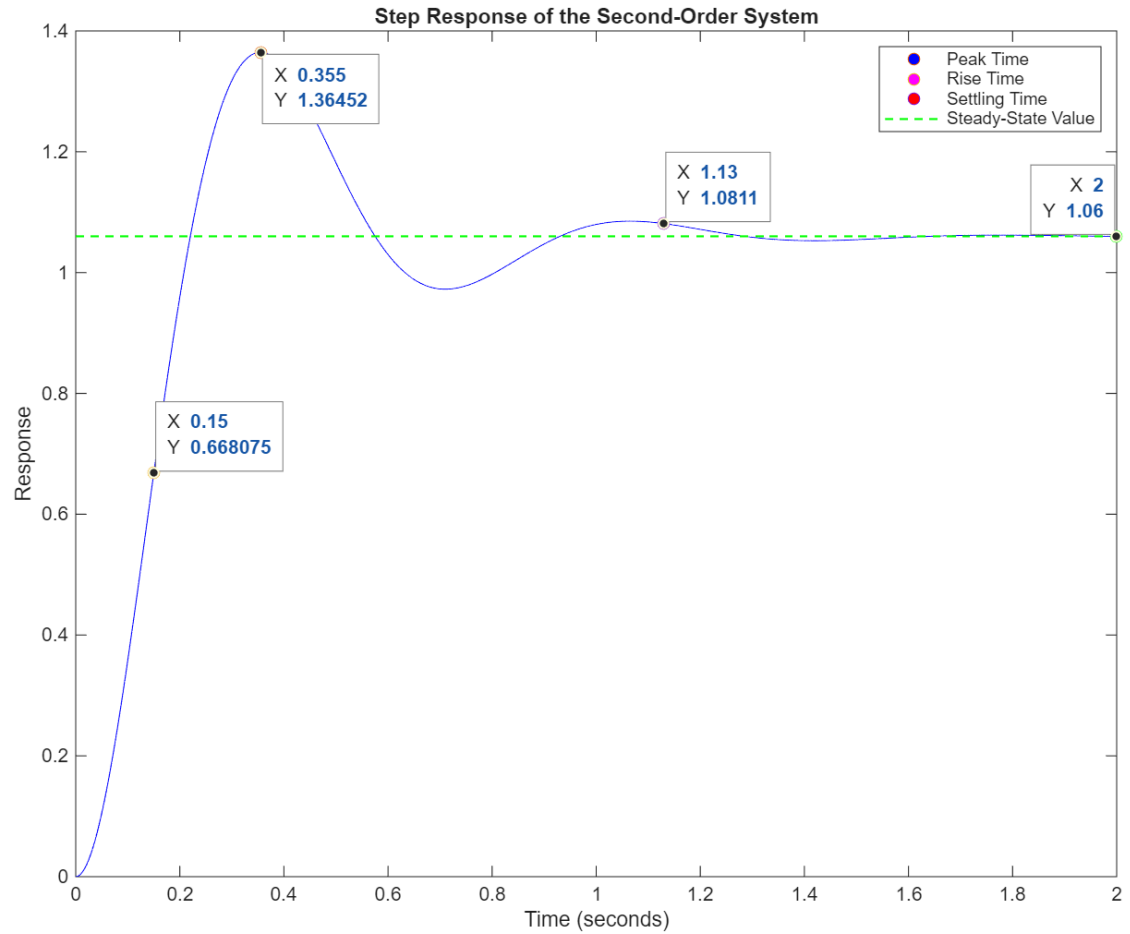
Post Lab Assignment 2)

$$\begin{aligned} \%OS &= 100 e^{-\frac{\xi \pi}{\sqrt{1-\xi^2}}} \\ \ln(0.287) &= -\frac{\xi \pi}{\sqrt{1-\xi^2}} \\ -1.251 &= -\frac{\xi \pi}{\sqrt{1-\xi^2}} \\ 1.251 &= \frac{\xi \pi}{\sqrt{1-\xi^2}} \\ \frac{1.251}{\pi} &= \frac{\xi}{\sqrt{1-\xi^2}} \\ 0.398^2 &= \frac{\xi^2}{1-\xi^2} \\ 0.1586(1-\xi^2) &= \xi^2 \\ 1.1586 \xi^2 &= 0.1586 \\ \xi^2 &= \frac{0.1586}{1.1586} = 0.13689 \\ \sqrt{\xi^2} &= \sqrt{0.13689} \\ \xi &= 0.369 \end{aligned}$$

$$w_n = \frac{\pi}{T_p \sqrt{1-\xi^2}}$$

$$\begin{aligned} w_n &= \frac{\pi}{0.355 \sqrt{1-(0.369)^2}} \\ w_n &= 9.5215 \end{aligned}$$

$$\begin{aligned} G(s) &= \frac{K w_n^2}{s^2 + 2 \xi w_n s + w_n^2} \\ G(s) &= \frac{1.06 * (9.5215)^2}{s^2 + 2(0.369)(9.5215)s + (9.5215)^2} \\ G(s) &= \frac{96.0985}{s^2 + 7.027 s + 90.659} \end{aligned}$$



## % Question 2

% Define the transfer function

numerator = 96.0985;

denominator = [1, 7.027, 90.659];

system = tf(numerator, denominator);

% Mark steady-state value, peak time, rise time, and settling time

steady\_state\_value = 1.06;

peak\_time = 0.355;

rise\_time = 0.15;

settling\_time = 1.13;

% Plot the step response

t = 0:0.001:2;

[y, t] = step(system, t);

figure;

plot(t, y, 'b-', 'MarkerSize', 1, 'DisplayName', 'Step Response');

hold on;

% Key point markers

h\_peak = plot(t(find(t >= peak\_time, 1)), y(find(t >= peak\_time, 1)), 'o', 'MarkerFaceColor', 'blue', 'DisplayName', 'Peak Time');

h\_rise = plot(t(find(t >= rise\_time, 1)), y(find(t >= rise\_time, 1)), 'o', 'MarkerFaceColor', 'magenta', 'DisplayName', 'Rise Time');

h\_settle = plot(t(find(t >= settling\_time, 1)), y(find(t >= settling\_time, 1)), 'o', 'MarkerFaceColor', 'red', 'DisplayName', 'Settling Time');

% Green dashed line at steady-state value

h1 = plot([0 max(t)], [steady\_state\_value steady\_state\_value], 'g--', 'LineWidth', 1);

% Green steady-state marker

h2 = plot(t(end), steady\_state\_value, 'go', 'MarkerFaceColor', 'green');

% --- Combine steady-state line + marker into ONE legend entry ---

h\_ss = [h1 h2];

% Labels

title('Step Response of the Second-Order System'); xlabel('Time (seconds)'); ylabel('Response');

% Final legend call (clean)

legend([h\_peak h\_rise h\_settle h\_ss], {'Peak Time', 'Rise Time', 'Settling Time', 'Steady-State Value'});